Production II: Cost, and the Feedback

Advanced Microeconomics 1: Economic Decision Making. Lecture 6

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Today's plan

1. The cost minimization problem — which we have already solved...
2. From $c(w, q)$ to supply: cost curves, $p = MC$, etc.
3. Trace through the effects of an indexed wage rise.
4. What we don't discuss.

The cost minimization problem

The cost minimization problem

CMP

Given input prices $w \gg 0$ and required output q:

$$c(w, q) = \min_{z \geq 0} \; w \cdot z \quad \text{s.t.} \quad f(z) \geq q,$$

with solution $z(w, q)$, the **conditional factor demand**.

- "Conditional": demand for inputs *conditional on producing* q. The scale decision is switched off; only the input-mix decision remains.
- Minimise expenditure subject to hitting a target level of an increasing function. This is the EMP with new labels.

The twin table

	Consumer (L2–L3)	Firm (today)
problem	$\min p \cdot x$ s.t. $u(x) \geq u$	$\min w \cdot z$ s.t. $f(z) \geq q$
value	$e(p, u)$: cost of living	$c(w, q)$: cost of producing
solution	$h(p, u)$: Hicksian demand	$z(w, q)$: conditional demand
Shephard	$\nabla_p e = h$	$\nabla_w c = z$
curvature	e concave in p	c concave in w
derivative matrix	$D_p h$ sym NSD, $D_p h\, p = 0$	$D_w z$ sym NSD, $D_w z\, w = 0$
law	compensated law of demand	compensated law of input demand

- The main difference: u was invisible and ordinal; q is observable and cardinal.
- $\Rightarrow$ today we will go fast. You've seen it all before!

Solving it: Lagrange, and what λ is now

- Interior FOC: $w_\ell = \lambda \, \partial f / \partial z_\ell$ for used inputs, so for any two:

$$\text{MRTS}_{\ell k} = \frac{w_\ell}{w_k}$$

  the input-mix version of tangency.

- $\lambda = \partial c(w, q)/\partial q$ — the cost of one more unit of output. λ is marginal cost.

The 15% energy cut, as curvature

- Concavity of c in w says: when one input price rises, cost rises *less than linearly* — because the firm re-optimises the mix.

- As in the consumer section, with the signs flipped: the fixed basket over-compensates the household because it prices no substitution. Here: the fixed input mix *over-states* the cost of a shock because it prices no substitution.

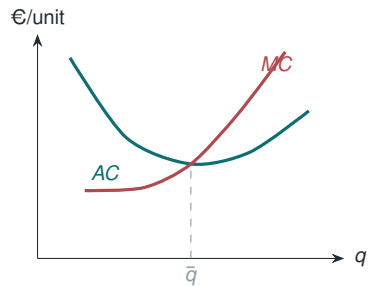
From cost to supply

From $c(w, q)$ to the cost curves

- Fix w; write $C(q) = c(w, q)$. Define

$$AC(q) = \frac{C(q)}{q}, \qquad MC(q) = C'(q).$$

- MC crosses AC at the minimum of AC: the **efficient scale** $\bar{q}$. (If AC is falling, the marginal unit is cheaper than the average, dragging the average down; and vice versa.)



NB: with CRS, $AC = MC =$ constant — the geometry degenerates.

Supply: $p = MC$, above AC

- Give the firm back its output choice, cost function in hand:

$$\max_{q \geq 0} \; p\,q - c(w, q) \qquad \text{FOC:} \quad p = MC(q)$$

- The competitive firm produces where price equals marginal cost — provided $p \geq AC$ at that output, otherwise it shuts down. The supply curve is the *MC* curve above *AC*.

- Anything that shifts *MC* shifts *supply*, and in the competitive benchmark, cost changes pass into prices through this equation.

The impact of inflation and indexation with the firm switched on

The setup

- Belgian wages are indexed. When the 2022 shock raised consumer prices, wages rose automatically.
- But the same event, from the firm's perspective: an economy-wide rise in w_L.
- Key point: Imported inflation. CPI includes import-inflation so wages indexation includes import inflation. Firm cost inflation necessarily doesn't. So if:
  1. Import prices rise faster than domestic prices, and...
  2. Consumption is more import-intensive than production, then...

  Wages will inflate faster than other inputs.

  So: indexation to the CPI insures workers against the domestic component of inflation and transfers the imported component between workers and firms.

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4. **Price**: in the competitive benchmark the industry supply shifts up, and the new equilibrium p is *higher*. Part of the wage rise returns to households as inflation.

From signs to sizes

- The five channels are signed. We want magnitudes. Use a small calibrated model of the average firm.
- Assumptions: **A1** CES cost function (the form the literature uses for this shock: Bachmann et al. 2022); **A2** constant returns; **A3** price taking; **A4** industry demand elasticity $\varepsilon_D = -1$.

$s_L = 0.11$	labour share of variable cost	108k Belgian firms	Bijnens et al.
$s_O = 0.89$	all other inputs	$= 1 - s_L$	—
$\sigma = 0.4$	substitution elasticity	from the employment evidence	BHMN back-out
$g_O = 5.9\%$	other input costs, 2022		—
g_O includes producer energy $+93\%$ (ECB) at its $\approx 3\%$ cost weight (EC energy report).			
$g_w = 9.6\%$	cheque $=$ 2022 avg CPI		Statbel

Channel 1

A1, written down, two inputs, labour L and everything else O:

$$c(w_L, w_O, q) \;=\; q\,\Phi(w)^{\frac{1}{1-\sigma}}, \qquad \Phi(w) \equiv a_L w_L^{1-\sigma} + a_O w_O^{1-\sigma}.$$

Take logs, differentiate with respect to $\ln w_i$:

$$\frac{\partial \ln c}{\partial \ln w_i} = \frac{1}{1-\sigma} \cdot \frac{(1-\sigma)\, a_i w_i^{1-\sigma}}{\Phi} = \frac{a_i w_i^{1-\sigma}}{\Phi} \overset{\text{Shephard}}{=} \frac{w_i z_i}{c} = s_i.$$

So, to a first-order:

$$\boxed{d \ln c \;=\; s_O\, g_O \;+\; s_L\, g_w \;=\; 5.2\% + 1.1\%}$$

Channel 1: costs rise by exposure $\times$ shock, before any behaviour.

Channel 4, the price

Under **A2** (CRS), cost is proportional to output: $c = q\,\varphi(w)$ with $\varphi(w) = c(w, 1)$. Hence

$$AC = MC = \varphi(w), \quad \text{flat in } q.$$

Under **A3**, an interior optimum requires $p = MC$; with $AC = MC$ this is also *zero profit*. This pins the price to input costs,

$$p_O = \varphi(w) \quad \Longrightarrow \quad \boxed{d \ln p_O = d \ln c = 5.2\% + 1.1\%}$$

Channel 4: the cost rise passes into the output price one-for-one — and the 1.1pp piece is the policy returning to households as inflation.

Channel 2

Shephard, computed on A1: $z_L = \partial c/\partial w_L = q\, a_L w_L^{-\sigma}\, \Phi^{\frac{\sigma}{1-\sigma}}$. Log-differentiate, using $\partial \ln \Phi / \partial \ln w_j = (1-\sigma) s_j$:

$$\frac{\partial \ln z_L}{\partial \ln w_L} = -\sigma(1 - s_L), \qquad \frac{\partial \ln z_L}{\partial \ln w_O} = +\sigma\, s_O = +\sigma(1 - s_L).$$

Equal and opposite — so only the *difference* of the shocks matters:

$$\boxed{d \ln z_L\big|_q \;=\; -\sigma\,(1 - s_L)\,(g_w - g_O) \;=\; -0.4 \times 0.89 \times 3.7\text{pp} = -1.3\%}$$

- **Why does the firm adjust labour at all?** If wages and other inputs rose *uniformly*, $g_w = g_O$, the box says zero: z is h.d.0 in w — the firm-side no-money-illusion theorem.
- The mix moves only because indexation made labour *relatively* more expensive: the CPI contains imported prices.

Channel 3

Unconditional labour is $L = z_L(w, q^*)$; under A2, $z_L \propto q$, so

$$d\ln L = d\ln z_L\big|_q + d\ln q^*, \qquad \text{and } \mathbf{A4} \text{ closes output: } \quad d\ln q^* = \varepsilon_D\, d\ln p_O.$$

$$d\ln L \;=\; \underbrace{-\sigma(1-s_L)(g_w - g_O)}_{\text{mix: } -1.3\%} \;+\; \underbrace{\varepsilon_D\,(s_O g_O + s_L g_w)}_{\text{scale: } -6.3\%} \;=\; -7.6\%$$

- Channel 3: more expensive output $\Rightarrow$ less sold $\Rightarrow$ less of *every* input, labour twice over.

Channel 3. Uses: CRS; A4. Assumes: partial equilibrium — the GE frame brackets it.

The firms reaction

	2022 cost shock	shock + indexation	the policy's addition
other-goods prices p_O	$+5.2\%$	$+6.3\%$	$+\mathbf{1.1}\%$
employment	-3.1%	-7.6%	$-\mathbf{4.5}\%$

- Wage indexation exacerbated things from the firms' side.
- Perhaps unexpectedly the main incidence lies on employment.

Before going back to households: one inconsistency to fix

- Block 1 priced only *energy*; but the cheque is indexed to *everything*.
- The fix: let the other price move too. Define π_O so the *average* basket reproduces observed 2022 inflation:

$$0.064 \times 57.9\% \;+\; 0.936\,\pi_O \;=\; 9.6\% \quad\Longrightarrow\quad \pi_O = 6.3\%.$$

- And the model just told us where π_O comes from:

$$\underbrace{6.3\%}_{\pi_O} = \underbrace{5.2\%}_{\substack{\text{other input costs through firms, } s_O g_O \\ \text{(incl. producer energy } +93\% \text{ at } \approx 3\% \text{ weight)}}} + \underbrace{1.1\%}_{\substack{\text{the policy itself} \\ s_L g_w}}$$

- So block 2 has opened block 1's black box: part of "other-goods inflation" was the energy shock arriving through firms — and part was the repair arriving as disease. Without the policy, π_O would have been 5.2%.

The household's bill, at full prices

Assume households have Stone–Geary preferences again.

	full bill	cheque (= avg CPI)	net
poor, $s_0 = 0.10$	11.3%	9.6%	−1.7%
rich, $s_0 = 0.06$	9.3%	9.6%	+0.3%

Same result as in lecture 4 but now looking at overall, not just energy, inflation. Proof → p. 31

Putting everything together

	no indexation	**automatic indexation**
no firms	(a) real wage: poor $-\mathbf{11.3}\%$, rich $-\mathbf{9.3}\%$ (b) employment: unchanged	(a) real wage: poor $-\mathbf{1.7}\%$, rich $+\mathbf{0.3}\%$ (b) employment: unchanged
with firms	(a) real wage: poor $-\mathbf{10.4}\%$, rich $-\mathbf{8.3}\%$ (b) employment: $-\mathbf{3.1}\%$	(a) real wage: poor $-\mathbf{1.7}\%$, rich $+\mathbf{0.3}\%$ (b) employment: $-\mathbf{7.6}\%$

The bottom line

- Allowing firms to adjust reveals the real losers from both inflation and indexation $\rightarrow$ those who became unemployed.
- The policy does a fairly good job of sheltering the remaining employed.
- Two things we haven't (and won't) discuss:
  1. Indexation has to be paid for — taxation.
  2. Those extra wages are spent — GE loop back to demand.

Why indexation fails, on both sides of the market

- **On the uniform component, it does not fail.** If every price and the wage rose by the same π, households are exactly compensated (*h* h.d.0 — L1) and firms change nothing (*z* h.d.0 — today). A perfect hedge against uniform inflation.

- **Consumer side (L4): a measurement failure.** The cheque prices no substitution, and one rate meets many baskets: $\pm$1–2pp across households.

- **Firm side (today): an incidence failure.** The CPI contains imported prices firms never receive, so indexation raises labour's *relative* price — the 3.7pp wedge — and so unemployment rises.

A reality check

- The model implies employment fell 7.6%! Did it? No — it actually rose.
- What went wrong with our modeling?
  1. Other things happened — post-COVID recovery.
  2. GE — indexation boosts demand as the money is spent, we need to endogenise demand.
- Real-life is (even more) complex!

What we have yet to consider

- But there's more!
- Thus far we have only considered things from a static perspective.
- The cost of inflation is not just price rises today, but the threat of future rises.
- Wage indexation insures (the employed) against these rises. What is the insurance value of the policy?
- To answer this we need to consider, choice under uncertainty.

Appendix

Derivation: the two-price Stone–Geary bill

Stone–Geary, $u = \beta \ln(x_E - \gamma) + (1-\beta)\ln x_O$. Its expenditure function (solve the EMP; or integrate Shephard twice):

$$e(p_E, p_O, u) \;=\; p_E\,\gamma \;+\; \kappa\, p_E^{\beta}\, p_O^{1-\beta}\, e^{u}, \qquad \kappa = \beta^{-\beta}(1-\beta)^{-(1-\beta)}.$$

Evaluate at u^0 using $e(p^0, u^0) = w$: the supernumerary part is worth $w - p_E^0\gamma$ at old prices, and it scales by the price ratios:

$$e(p^1, u^0) = p_E^1\gamma + (w - p_E^0\gamma)\left(\frac{p_E^1}{p_E^0}\right)^{\beta}\left(\frac{p_O^1}{p_O^0}\right)^{1-\beta}$$

Subtract w, divide by w, write $\tilde{\gamma} = p_E^0\gamma/w$:

$$\frac{-CV}{w} = \tilde{\gamma}\, g_E + (1-\tilde{\gamma})\Big[(1+g_E)^{\beta}(1+\pi_O)^{1-\beta} - 1\Big].$$

Set $\pi_O = 0$: L4's one-price formula, exactly. The subsistence quantity γ is priced at the *full* energy rise; only the flexible remainder substitutes.

← Back to p. 24